

Utkarsh Singh

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Professional Summary

Computer Science undergraduate (9.34 CGPA) specializing in Applied AI and Machine Learning. My work spans Generative AI, NLP, Deep Learning and predictive modeling, building intelligent systems for healthcare, scientific discovery, and information retrieval. Seeking AI/ML Engineer or Data Science opportunities.

Education

Bachelor of Technology, Computer Science and Business Systems	July 2023 – June 2027 (present)
Bharati Vidyapeeth Deemed University, Pune	CGPA: 9.36
Higher Secondary (Class XII): PCM & Psychology	April 2015 – March 2023
City Montessori School, Lucknow	XII: 94% X: 91.8%

Skills

Languages: Python, SQL, JavaScript, TypeScript

Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, Hugging Face Transformers, SHAP, ChromaDB, SMOTE

Web & Databases: React, Next.js, Node.js, FastAPI, PostgreSQL, Supabase, Prisma ORM

Tools: Git, GitHub, VS Code, Streamlit Cloud

Certifications: OpenAI GPT-4, Intro to Data Science, Information Security, NLP: Infosys Springboard; Python Programming: Coursera

Projects

Sinter(AI Perspective & Counter-Argument Engine) | Github March 2026–May 2026
Node.js, React, PostgreSQL (Supabase), Llama-3, RAG Architecture

- Architected a **RAG** Chrome extension fetching **5** sources/query, deduplicating into **8** unique snippets; replaced hallucinated LLM citations with live **Serper API** retrieval, guaranteeing **100%** verifiable source URLs.
- Integrated **Llama-3.3-70B** as the inference engine within a **RAG pipeline**, generating three evidence-grounded counter-arguments per article from a **5,000-character** retrieved context.
- Designed a **PostgreSQL** “Steel-man vs. Straw-man” crowdsourced arbitration system with a caching layer, reducing query latency from **6s** to **<100ms** for repeat article analyses.

exHUMA:AI Exoplanet Detector | Deployment Link January 2026–February 2026
TensorFlow, CNNs, SHAP, SMOTE

- Engineered a Multi-Branch **CNN-FFT** pipeline ranking **5,087 stars** into **Top-20 candidates**, retrieving 4/5 confirmed exoplanets (**Recall@20: 80%**).
- Eliminated false positives via automated **BLS Phase Folding** and **Vetting SNR filtering**; applied **SMOTE** with custom class weighting to isolate faint transits against an extreme **<1%** positive sample rate.
- Integrated **SHAP** with a custom reshaping wrapper to generate verifiable transit-dip **XAI heatmaps**, deploying a scientific dashboard that improves manual survey efficiency by **80%**.

Manas Mitra:Multilingual Student Support Chatbot | Collaborative | Deployment Link December 2025
Next.js, FastAPI, Gemini API, RAG (ChromaDB), SentenceTransformers

- Architected a clinical **RAG pipeline** using **ChromaDB** to map user queries to **25 distinct CBT frameworks** via **Gemini 2.0**.
- ”Engineered a zero-latency **semantic routing layer** in **Next.js**, acting as a circuit breaker to instantly route acute distress queries to national helplines while bypassing LLM API dependencies.

Leadership & Achievements

- Organized a **200+** participant prompt engineering competition at Bharatiyam 2026 as **CSBS Deputy Joint Secretary**.
- Secured **Top 30** out of **12000+** teams in **Innervive 9** National Hackathon in the track of AI/ML.